Discrete Mathematics_2023_PYQ

Group-A (Very Short Answer Type Question)

Answer any ten of the following: [1 x 10 = 10]

1. What is a cycle in a graph?
2. Which law of inference states that 'if A implies B, and B implies C, then A implies C'?
3. What do you mean by the symmetric difference of set A and B?
4. How many reflexive relations are possible on a set with n elements?
5. What do you mean by a 'relation'?
6. What is a spanning tree in a graph?
7. Write the Absorption laws in respect of SET Theory.
8. Write the contrapositive of $\sim p \rightarrow \sim q$.
9. What is the inverse of $p \rightarrow \sim q$?
10. 'K6 is a planar graph.' – State TRUE or FALSE.
11. What is Cantor's diagonal argument?
12. Write the De Morgan's laws of SET operations.

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Group-B (Short Answer Type Question)

Answer any three of the following: [5 x 3 = 15]

2. What is an inverse function? How do you obtain inverse of a function? Explain with example. [5]
3. Five speakers A, B, C, D and E speak in a meeting one after the other. Find the probability that A speaks before B. [5]
4. Prove by induction: $n^2 + n$ is even where n is a positive integer. [5]
5. There are 5 white balls, 4 red balls, and 3 blue balls in a box. If you draw 2 balls at random from the box, what is the probability that both balls are either white or red? [5]
6. In how many ways you invite one or more of your five friends to your birthday party? [5]

Group-C (Long Answer Type Question)

Answer any three of the following: [15 x 3 = 45]

7. (a) Define a SET with proper examples according to the classical set theory. What are the different set operations usually performed – explain with examples. [7]

(b) What do you mean by finite and infinite sets? Also, explain the concepts of finite countable sets, infinite countable sets and infinite uncountable sets with proper examples. [2+6]

8. (a) How many non-negative integral solutions are there to the equation

$$x_1 + x_2 + x_3 + x_4 = 20?$$ [7]

(b) Show that number of prime numbers is infinite. [8]

9. (a) Show that universal quantifier distributes over conjunction and existential quantifier distributes over disjunction. [8]

(b) Brown, Jones and Smith are suspected of income tax evasion. They testify under oath as follows – Brown: Jones is guilty and Smith is innocent; Jones: If Brown is guilty, then so is Smith; Smith: I am innocent, but at least one of the others is guilty. Assuming everyone told the truth. Who is/are guilty/innocent? [7]

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10. (a) What do you mean by Eulerian and Hamiltonian walks? [5]

(b) What is Vertex Colouring and Colouring of Edges in graph theory? [5]

(c) Define Minimal Spanning tree (MST) with an example. [5]

11. (a) What is a bipartite graph? How do you determine if a graph is bipartite or not? [5]

(b) Show that trees are bipartite graphs. [5]

(c) Determine the chromatic polynomial of K_n. [5]